

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

This observing program is scientifically interesting as finding exoplanets can help us understand the planet diversity in our universe, understand how planetary systems form, and whether life can be supported on planets outside of Earth and the solar system. Transiting exoplanets can allow us to measure their precise mass and radius, which are very useful measurements for this.

For this project, we are trying to observe TOI 1516b using the transit method of detecting exoplanets. Through this, we can find out what the mass and radius of the exoplanet is, and using that, we can determine if the target is a binary-star or an exoplanet.

Telescope Requested (0.5 points): Kuiper 61” telescope on Mt. Bigelow

Instrument (0.5 points): Mont 4K

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

As the host star of our target is quite bright, the moon brightness acceptable for our program is about 60-70%. This is because our target is more than 5 hrs of RA away from the moon on our observing night on 10/02. The moon can impact the sensitivity of our observations as it can increase the brightness of the sky in the background, increasing the noise of our data.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example “Object can be observed Feb-March.”

Object Name	Right Ascension	Declination	V magnitude	Object Type	Visibility
TOI 1516b	22 40 20.26	+69 30 13.455	10.9	Exo-Planet	October 2 nd , 2026
HD 209458b	22 03 10.77	+18 53 03.549	7.7	Exo-Planet	October 3 rd , 2026

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Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? **(2 points)**

The exposure time required for the project is around 5 seconds, and the signal-to-noise necessary for the observations is around 1000.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? **(2 points)**

Through our observations, we can see how much of the host star's light the target is blocking during its transit, which we can plot on a light-curve. Using this, we can estimate the target's radius and mass, allowing us to confirm whether it's an exoplanet or a companion star.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

In case of poor weather, we can use archival data of two transiting planets from the planetary system of the star TOI-1458, specifically from the discovery paper "**TOI-1468: A system of two transiting planets, a super-Earth and a mini-Neptune, on opposite sides of the radius valley**" by P. Chaturvedi et al. Using their measurements and light curves, we can determine the mass and radius of one or both of the exoplanets.